Hardness test — data sheet

UMT 312L · Group 1 (Friday), handout **Gp 4** · Mayank Bohat, Chidambar · 13 November 2026

$HV = 1.8544\,F/d^2$ $HB = 2F/\pi D\left(D - \sqrt{D^2 - d^2}\right)$, $D = 2.5$ mm F in kgf, d in mm · —— = not measured

1. Material type (Brinell and Vickers)

MS, Cu, SS · Brinell 2.5 mm ball at 187.5 kgf and Vickers at 100 kgf · five indents each

Mild steel

#	**Brinell**, 187.5 kgf d_1 (mm)	d_2 (mm)	mean d (mm)	HB	**Vickers**, 100 kgf d_1 (mm)	d_2 (mm)	mean d (mm)	HV
1	1.050	1.030	1.0400	210.7	——	——	——	——
2	1.027	1.038	1.0325	213.9	——	——	——	——
3	1.044	1.021	1.0325	213.9	——	——	——	——
4	——	——	——	——	——	——	——	——
5	——	——	——	——	——	——	——	——
mean				**212.9**				——
95 %				± 4.6				——

Stainless steel

#	**Brinell**, 187.5 kgf d_1 (mm)	d_2 (mm)	mean d (mm)	HB	**Vickers**, 100 kgf d_1 (mm)	d_2 (mm)	mean d (mm)	HV
1	0.873	0.869	0.8710	304.8	0.799	0.771	0.7850	300.9
2	0.879	0.882	0.8805	298.1	0.776	0.766	0.7710	312.0
3	0.853	0.862	0.8575	314.8	0.791	0.770	0.7805	304.4
4	——	——	——	——	——	——	——	——
5	——	——	——	——	——	——	——	——
mean				**305.9**				**305.8**
95 %				± 20.9				± 14.0

Copper

#	**Brinell**, 187.5 kgf d_1 (mm)	d_2 (mm)	mean d (mm)	HB	**Vickers**, 100 kgf d_1 (mm)	d_2 (mm)	mean d (mm)	HV
1	1.474	1.466	1.4700	99.9	1.377	1.328	1.3525	101.4
2	1.459	1.470	1.4645	100.8	1.352	1.349	1.3505	101.7
3	1.462	1.480	1.4710	99.8	1.386	1.337	1.3615	100.0
4	——	——	——	——	——	——	——	——
5	——	——	——	——	——	——	——	——
mean				**100.1**				**101.0**
95 %				± 1.3				± 2.2

2. Effect of load (MS, Brinell)

mild steel, Brinell 2.5 mm ball · three indents at each load

Load (kgf)	**21.2**	**60**	**100**	**150**	**187.5**
indent 1, d_1/d_2 (mm)	0.415/0.442	0.601/0.608	0.754/0.760	0.939/0.929	1.039/1.038
indent 2, d_1/d_2 (mm)	0.444/0.443	0.588/0.606	0.797/0.768	0.931/0.946	1.035/1.031
indent 3, d_1/d_2 (mm)	0.448/0.447	0.600/0.604	0.729/0.730	0.935/0.933	1.042/1.035
mean d (mm)	0.4398	0.6012	0.7563	0.9355	1.0367
projected area $\pi d^2/4$ (mm^2)	0.1519	0.2838	0.4493	0.6873	0.8440
HB	**138.4**	**208.3**	**217.4**	**210.3**	**212.1**

3. Effect of heat treatment (Al–Cu, Vickers)

Vickers at 100 kgf · five indents on each condition

As-solutionized, 550 °C 1 h, water quenched

#	d_1 (mm)	d_2 (mm)	mean d (mm)	HV
1	1.176	1.236	1.2060	127.5
2	1.234	1.187	1.2105	126.6
3	1.245	1.178	1.2115	126.3
4	1.249	1.202	1.2255	123.5
5	1.247	1.189	1.2180	125.0
mean				**125.9**
95 %				± 1.5

Aged 230 °C, 60 min

#	d_1 (mm)	d_2 (mm)	mean d (mm)	HV
1				
2				
3				
4				
5				
mean				
95 %				

Aged 230 °C, 30 min

#	d_1 (mm)	d_2 (mm)	mean d (mm)	HV
1				
2				
3				
4				
5				
mean				
95 %				

*As-solutionized: 550 °C for 1 hr, then water quenched.